

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 222 — Past Questions & Materials
by Renance

Course: GET 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — Information technology in engineering GET 222 2017/2018

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester

SESSION: 2017/2018



COURSE CODE: GET 222

COURSE TITLE: Information Technology in Engineering

DATE: 17/09/2018

TIME ALLOWED: 2Hrs

Instruction: Answer Questions ONE and Any other THREE

QUESTION ONE

- (1a) In the industrial automation field, analog signals are still distributed using the venerable 4mA to 20mA current loop. In this technique, the output from any transducer is conditioned by means of some electronics to generate a current of 4mA at the bottom of the scale and vary continuously up to 20mA at the top of the scale. The block diagram in Figure 1 describes just such an application where a 0Vdc to 10Vdc input signal is translated to 4-20mA output. The transfer function is $I_{out} = ((V_{in}/V_{fullscale}) * 16) + 4$. The current, I_{out} , is measured in milliamps. Generate the table for the relationship between the input voltage and the output current using traditional spreadsheet. (4Mks)
- (1b) from the table in (1a) explain how to plot 'Scatter with Straight lines and Markers' chart to analyze the linearity of the system using Microsoft excel 2016. (5Mks)

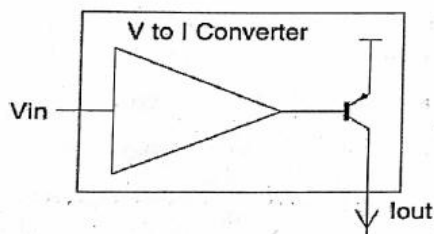


Figure 1 0Vdc to +10Vdc is converted to 4-20mA through this module

- (1c) Explain Database Schema. (5Mks)
- (1d) State the functions performed by the following shortcut keys: (i) CTRL+Shift+G (ii) CTRL+Shift+I (iii) CTRL+Alt+K (iv) CTRL+Shift+C (v) CTRL+Shift+U (vi) CTRL+Alt+W (vii) Alt+Spacebar (viii) Shift+WindowsKey+M (ix) Alt+Shift+F1 (x) Alt+Shift+T (10Mks)
- (1e) Name and explain the parts labelled in Figure 2. (10Mks)

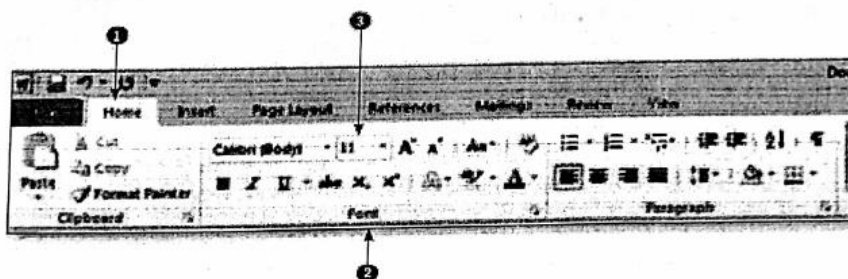


Figure 2

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QUESTION TWO

- (2a) (i) What is a Microsoft PowerPoint icon? (ii) List four different elements you can use or add while creating a Microsoft PowerPoint presentation. (3Mks)
- (2b) (i) Differentiate between transition and animation in Microsoft PowerPoint. (ii) List the major views in Microsoft PowerPoint. (4Mks)
- (2c) Outline the five basic steps you can take to create a Microsoft PowerPoint presentation on the Role of Information Technology in Combating Terrorism. (5Mks)

QUESTION THREE

- (3a) A university is preparing admission letters for it's over 5000 newly admitted students. The letters should contain each student's address, phone number and name. Outline three basic steps in doing this in MS-Word (2016). (3Mks)
- (3b) How do you write this expression, $\int_1^2 8x^2 + \frac{1}{2}x \, dx$ using MS Word (2016)? (4Mks)
- (3c) List any five (5) items found in the File tab in MS Word (2016) (5Mks)

QUESTION FOUR

- (4a) Define database management system? (2Mks)
- (4b) List any eight applications of DBMS. (5Mks)
- (4c) What are the advantages of using a DBMS? (5Mks)

QUESTION FIVE

Represent the student data below in a MS Excel sheet and then write one instance of formulae and or functions to extract the following information:

- (5a) Complete the columns for total score (1 Mk)
- (5b) Complete the column for grades as follows (i) 70 and above = distinction, (ii) 60 to 69 = merit, (iii) 40 to 59 = pass, (iv) less than 40 = fail. (2 Mks)
- (5c) Compute number of students with score equal to and greater than 50. (1 Mk)
- (5d) Compute number of students who scored between 45 and 65. (2 Mks)
- (5e) Compute the mean score of all the students. (1 Mk)
- (5f) Write a formula to identify the highest score. (1 Mk)
- (5g) Write a formula to identify the lowest score. (1 Mk)
- (5h) Write a formula to assign positions to the students according to their scores. Where the first position goes to the person with the highest score (1 Mk)
- (5i) Draw and label an actual bar chart to show the distribution of student scores according to the categories in (b) above (1 Mk)
- (5j) Draw and label an actual pie chart to show the distribution of student scores according to the categories in (b) above (1 Mk)

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Table 1

	C	D	E	F	G	H	I
	Mat No	Name	Test (30%)	Exam (70%)	Total (100%)	Grade	Positions
1	1	Alex John	12	25			
2	2	Betty Grace	5	52			
3	3	Ufoma Elohor	16	34			
4	4	Christen James	17	65			
5	5	Christy Peace	10	33			
6	6	Akpos Obukowho	14	27			
7	7	James Bond	20	46			
8	8	Ajiri Uyouyo	3	16			
9	9	Joseph Chukwudi	9	52			
10	10	Tracy John	27	60			

QUESTION SIX

State the functions of the parts labelled in Figure 3

(12Mks)

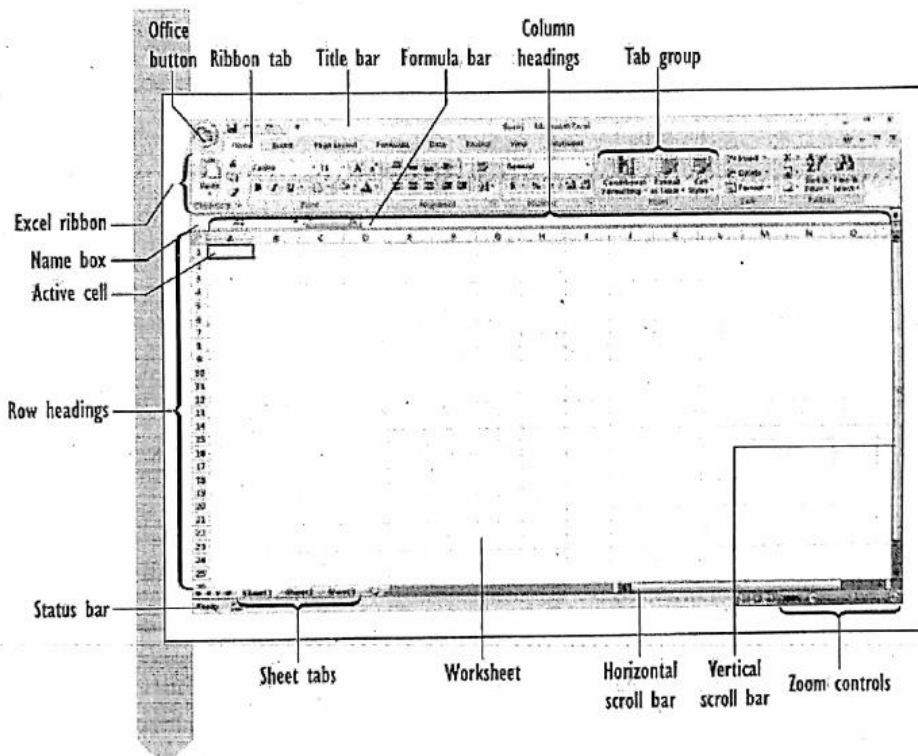


Figure 3

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester
SESSION: 2018/2019



E CODE: GET 212

COURSE TITLE: Computer
Programming

DATE: 11/05/2019

TIME ALLOWED: 2Hrs

Instruction: Answer FOUR Questions

QUESTION ONE

Write a complete program in C++ to solve the quadratic equation $ax^2 + bx + c = 0$; $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Be sure that the program can handle the special cases; where $a = 0$, where the two roots are equal, and where the discriminant $b^2 - 4ac$ is negative. (17.5 marks)

QUESTION TWO

- (a) Describe the following commands in MATLAB %, length, clc, diary, and clear.
- (b) In tabular form state the difference between script files and function files.
- (c) What is a Computer Language, state and explain the three categories. (17.5 marks)

QUESTION THREE

- (a) In tabular form differentiate between main memory (RAM) and secondary memory (Hard drive).
- (b) State five major operations or functions of a computer.
- (c) State five characteristics of a computers that have made them so powerful and universally useful. (17.5 marks)

QUESTION FOUR

Write a MATLAB function to obtain the roots of the quadratic equation $ax^2 + bx + c = 0$ (17.5marks)

QUESTION FIVE

- (a) Explain with examples the difference between system software and application software of a computer.
- (b) State and explain the three main parts of a central processing unit (CPU) of a computer.
- (c) State the three types of secondary memory of a computer. (17.5 marks)

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QUESTION SIX

For the circuit of figure Q6 shown below, write a MATLAB script to find the nodal voltages V_1 , V_2 and V_3 . (17.5 marks)

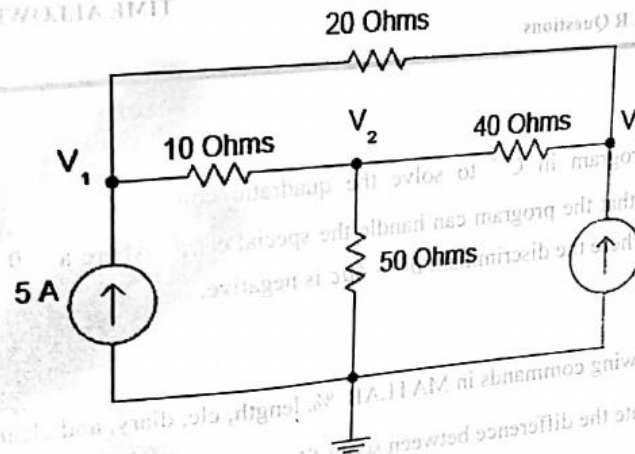


Figure Q6

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Paper 2 of 3 — GET 222 information technology in engineering 2017/2018

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester

SESSION: 2017/2018



COURSE CODE: GET 222
 COURSE TITLE: Information
 Technology in Engineering
 TIME ALLOWED: 2Hrs

DATE: 17/09/2018

Instruction: Answer Questions ONE and Any other THREE

QUESTION ONE

- (1a) In the industrial automation field, analog signals are still distributed using the venerable 4mA to 20mA current loop. In this technique, the output from any transducer is conditioned by means of some electronics to generate a current of 4mA at the bottom of the scale and vary continuously up to 20mA at the top of the scale. The block diagram in Figure 1 describes just such an application where a 0Vdc to 10Vdc input signal is translated to 4-20mA output. The transfer function is $I_{out} = ((V_{in}/V_{fullscale}) * 16) + 4$. The current, I_{out} , is measured in milliamps. Generate the table for the relationship between the input voltage and the output current using traditional spreadsheet. (4Mks)
- (1b) from the table in (1a) explain how to plot 'Scatter with Straight lines and Markers' chart to analyze the linearity of the system using Microsoft excel 2016. (5Mks)

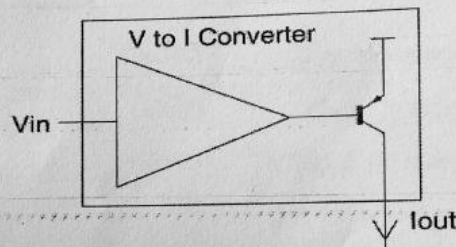


Figure 1 0Vdc to +10Vdc is converted to 4-20mA through this module

- (1c) Explain Database Schema. (5Mks)
- (1d) State the functions performed by the following shortcut keys: (i) CTRL+Shift+G (ii) CTRL+Shift+I (iii) CTRL+Alt+K (iv) CTRL+Shift+C (v) CTRL+Shift+U (vi) CTRL+Alt+W (vii) Alt+Spacebar (viii) Shift+WindowsKey+M (ix) Alt+Shift+F1 (x) Alt+Shift+T (10Mks)
- (1e) Name and explain the parts labelled in Figure 2. (10Mks)

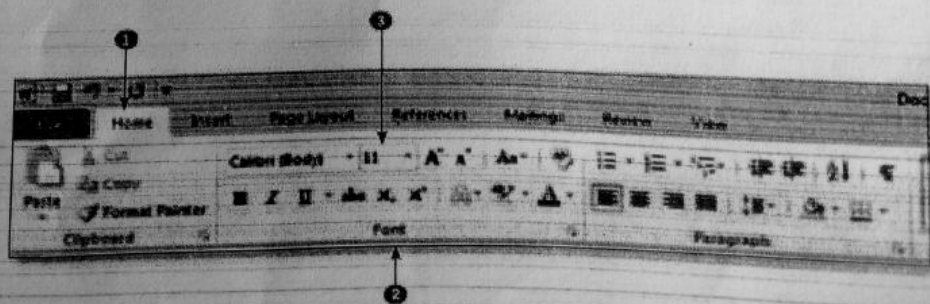


Figure 2

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QUESTION TWO

(2a) (i) What is a Microsoft PowerPoint icon? (ii) List four different elements you can use or add while creating a Microsoft PowerPoint presentation. (3Mks)

(2b) (i) Differentiate between transition and animation in Microsoft PowerPoint. (ii) List the major views in Microsoft PowerPoint. (4Mks)

(2c) Outline the five basic steps you can take to create a Microsoft PowerPoint presentation on the Role of Information Technology in Combating Terrorism. (5Mks)

QUESTION THREE

(3a) A university is preparing admission letters for its over 5000 newly admitted students. The letters should contain each student's address, phone number and name. Outline three basic steps in doing this in MS-Word (2016). (3Mks)

(3b) How do you write this expression, $\int_1^2 8x^2 + \frac{1}{2}x \, dx$ using MS Word (2016)? (4Mks)

(3c) List any five (5) items found in the File tab in MS Word (2016) (5Mks)

QUESTION FOUR

(4a) Define database management system? (2Mks)

(4b) List any eight applications of DBMS. (5Mks)

(4c) What are the advantages of using a DBMS? (5Mks)

QUESTION FIVE

Represent the student data below in a MS Excel sheet and then write one instance of formulae and or functions to extract the following information:

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(5c) Compute number of students with score equal to and greater than 50. (1 Mk)

(5d) Compute number of students who scored between 45 and 65. (2 Mks)

(5e) Compute the mean score of all the students. (1 Mk)

(5f) Write a formula to identify the highest score. (1 Mk)

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QUESTION SIX

(12Mks)

State the functions of the parts labelled in Figure 3

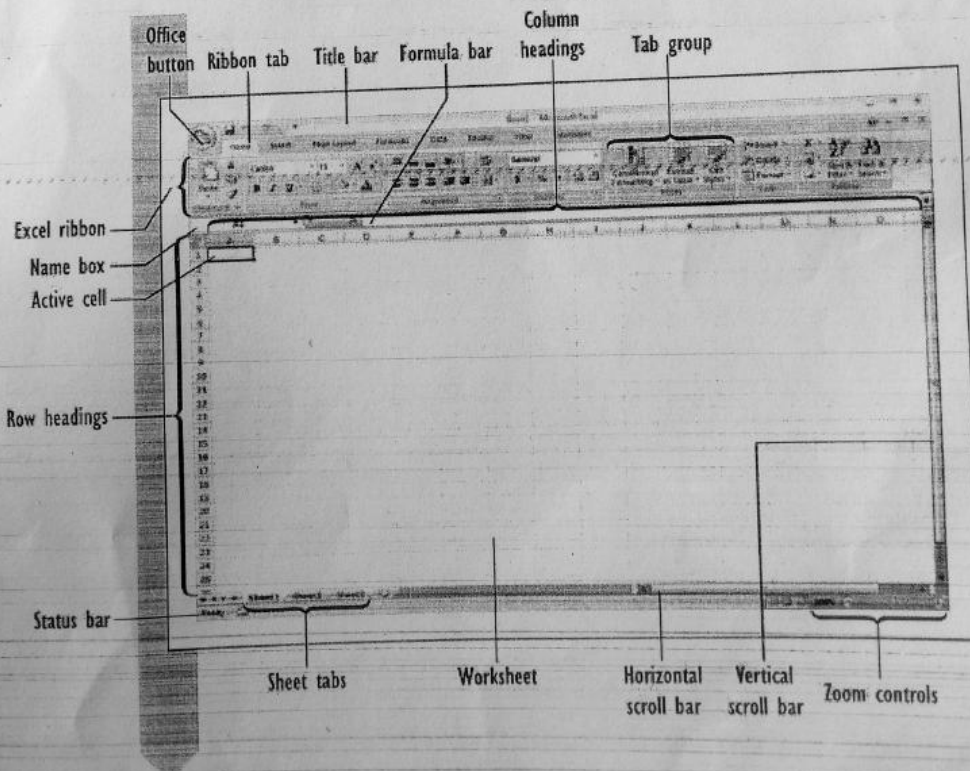


Figure 3

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Paper 3 of 3 — GET 222 PAST QUESTIONS

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester

SESSION: 2017/2018

COURSE CODE: GET 222

COURSE TITLE: Information

Technology in Engineering

DATE: 17/09/2018 TIME ALLOWED: 2Hrs

Instruction: Answer Questions ONE and Any other THREE

QUESTION ONE

(1a) In the industrial automation field, analog signals are still distributed using the venerable 4mA to 20mA current loop. In this technique, the output from any transducer is conditioned by means of some electronics to generate a current of 4mA at the bottom of the scale and vary continuously up to 20mA at the top of the scale. The block diagram in Figure 1 describes just such an application where a 0Vdc to 10Vdc input signal is translated to 4-20mA output. The transfer function is $T_{out} = ((V_{in}/V_{punscale}) * 16) + 4$. The current, I_{out} , is measured in milliamps. Generate the table for the relationship between the input voltage and the output current using traditional spreadsheet. (4Mks)

(1b) from the table in (1a) explain how to plot 'Scatter with Straight lines and Markers' chart to analyze the linearity of the system using Microsoft excel 2016. (5Mks)

V to I Converter

vin 4 t I

V Iout .

Figure 1 0Vdc to +10Vdc is converted to 4-20mA through this module

(1c) Explain Database Schema. (5Mks)

(1d) State the functions performed by the following shortcut keys: (i) CTRL+Shift+G (ii) CTRL+Shift+ (iii) CTRL+Alt+K (iv) CTRL+Shift+C (v) CTRL+Shift+U (vi) CTRL+Alt+W (vii) Alt+Spacebar (viii) Shift WindowsKey+M (ix) Alt+Shift+F1 (x) Alt+Shift+T | (10Mks)

(1e) Name and explain the parts labelled in Figure 2. (10Mks)

QUESTION TWO

(2a) (i) What is a Microsoft PowerPoint icon? (ii) List four different elements you can use or add

Titles : (i) Major

(2b) (i) Differentiate between transition and animation in Microsoft PowerPoint. (ii) List the main views in Microsoft PowerPoint. (4Mks)

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QUESTION THREE

(3a) A university is preparing admission letters for it's over 5000 newly admitted students. The letters should contain each student's address, phone number and name. Outline three basic steps in doing this in MS-Word (2016). (3Mks)

(3b) How do you write this expression, $\frac{1}{2}x^2 + 3x - 5$ using MS Word (2016)? (4Mks)

(3c) List any five (5) items found in the File tab in MS Word (2016) (5Mks)

QUESTION FOUR

(4a) Define database management system? (2Mks)

(4b) List any eight applications of DBMS. (5Mks)

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Table 1

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5 4 Christen James 17 65

6 5 Christy Peace 10 33

7 6 Akpos Obukowho 14 27

8 7 James Bond 20 46

9 8 Ajiri Uyouyo 3 16

10 9 Joseph Chukwudi 9 52

11 10 Tracy John 27 60

QUESTION SIX

State the functions of the parts labelled in Figure 3 (12Mks)

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Figure 3

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE OF TECHNOLOGY

EXAMINATION: Second Semester é £ CODE: GET 212

SESSION: 2018/2019 5 _ 1] COURSE TITLE: Computer

re r > Programming

DATE: 11/05/2019 TIME ALLOWED: 2Hrs

Instruction: Answer FOUR Questions

QUESTION ONE

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$\sim b \pm \sqrt{b^2 - 4ac}$, .

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QUESTION FIVE \

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(a) Explain with examples the differenc
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(b) State and explain the three main parts of a central processing unit (CPU) of a computer.

(c) State the three types of secondary memory of a computer. (47.5 marks)

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QUESTION SIX

For the circuit of figure Q6''shown below, write a MATLAB 'script' to find' the 'nodal voltagesV,, V2 and V3. (17.5 marks)

5A

+ Figure Q6